

Leu Anthony

Passionate about systems combining software, electronics and robotics, I wish to contribute to multidisciplinary technical projects by participating in the design of software solutions for embedded systems

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Driving License B

Education

Engineer in Informatics

Specialisation embedded systems

ISIMA (Clermont Auvergne INP)

2025 Clermont-Ferrand

Preparatory class for the Grandes Écoles (CPGE)

Mathematics & Physics (MP)

2020-22 *Nîmes*

Scientific Baccalaureate

European English Section
Honors

2020 *Cavaillon*

Technical Skills

- **Languages:** C++, C, C#, Python
- **Development:** OOP, Git, Unity, Visual Studio, RobotStudio
- **Embedded systems:** Arduino, Microcontroller, Datasheet analysis, Sensor and actuator integration
- **Communication protocol:** TCP/IP, Wi-Fi, MQTT, I2C, SPI

Interpersonal skills

- Autonomy
- Adaptability
- Multidisciplinary teamwork
- Analysis and problem solving

Languages

- **French:** mother tongue
- **English:** Advanced Level C1
  TOEIC 960/990 (2024)

Leisure

- **Reading:** English novels on online platforms
- **Swimming:** 7 years of practice, including competitions before a leisure-oriented practice

Embedded Software Development Engineer

Experiences

Software Development & Robotics Engineer - Internship

Development of a pick-and-place robotic arm

Trinity College Dublin

April-August 2025 *Dublin, Ireland*

- Design of a **C++** application to manage the logic of the movement of pieces and the link with other modules including the **Stockfish chess engine**
- Development of a **RAPID** program on **RobotStudio** to control the ABB robotic arm
- Design of a clamp controlled by an **ESP32** programmed on **Arduino IDE** in collaboration with a mechanical profile
- Implementation of the TCP communication protocol via Wi-Fi and Ethernet between the PC application, the robotic arm and the gripper.
- Initiation of a **computer vision module** (**CNN**) in **Python** to train a model in piece recognition in order to make the robot autonomous.
- Drafting of technical documentation to facilitate the use, maintenance and resumption of the project.

Software Development & Embedded Systems Engineer - Intern

Design of a 3D environment reconstruction system

XAP-Technology

March-July 2024 *Calvisson*

- Development on **Arduino** (**C/C++**) to **integrate sensors and actuators** (LiDAR and servo motor)
- Design of a **Unity** application (**C#**) to visualize and exploit the data collected.
- Setting up data acquisition and transmission (**serial, MQTT)**
- Identification of the origin of electronic failures on a prototype by analysis of **PCB schematics**, **datasheets** and use of multimeter.
- Validation of the diagnosis in **collaboration** with an electronic expert and implementation of corrective actions with a technician
- **Analysis** of scientific publications to identify algorithms that meet the needs of the project
- **Comparison** of approaches and implementation of the most suitable algorithm according to the established criteria

Technical agent in a seaside resort

Le Barcarès Town Hall

Summer 2021 & 2023 *Le Barcarès*

- Ensure the maintenance and cleanliness of public spaces
- Educating the public about tourism and urban planning

Projects

End of second year project (≃ 60h)

Claret & BUT Industrial Engineering and Maintenance Group

February 2024 *Clermont-Ferrand*

- Integration of the control of a robot in augmented reality on **Unity** (**C#**) on a tablet in connection with Stäubli Robotics Suite in **real time**
- Creating a Robot Simulation Application
- Setting up **TCP communication**

End of first year project (≃ 30h)

ISIMA (Clermont Auvergne INP)

June 2023 *Clermont-Ferrand*

- Application of **machine learning** on a game built in **C** with **SDL2** in a group of three students